

Machine Learning Algorithms for Money Laundering Detection: A Comparative Performance Analysis

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ABSTRACT

Money laundering poses a severe threat to the integrity of financial systems globally, with Bangladesh's growing digital banking sector presenting unique detection challenges. This study implements and evaluates five machine learning (ML) algorithms — XGBoost (Gradient Boosting), K-Nearest Neighbours (KNN), Random Forest (RF), Isolation Forest, and Support Vector Machine (SVM) — on a synthetic dataset of 2,000 individual bank customer transaction records constructed to reflect Bangladesh retail banking patterns. The dataset comprises 17 features spanning KYC/CDD profile attributes, transactional behaviour, alert and case history, and engineered risk indicators, with a class distribution of 70% normal and 30% suspicious transactions. SMOTE was applied to the training set to address class imbalance. The study evaluates the accuracy and precision of the implemented algorithms alongside recall, F1-score, and AUC-ROC metrics. XGBoost achieved the highest performance with accuracy of 97.00%, precision of 94.26%, recall of 95.83%, F1-score of 95.04%, and AUC-ROC of 0.9971. Statistical validation through 10-fold cross-validation and paired t-tests confirmed XGBoost's superiority over KNN ($p < 0.001$), Random Forest ($p = 0.0009$), and SVM ($p < 0.001$). Feature importance analysis identified `Previously_SAR_Filed`, `Internal_Risk_Rating`, and `Previous_Risk_Event` as the most predictive variables. Results are specific to the synthetic dataset used; generalisation requires independent validation on real banking data.

Keywords: Machine learning; money laundering detection; XGBoost; Random Forest; SVM; KNN; Isolation Forest; synthetic dataset; Bangladesh banking sector; SMOTE; comparative performance analysis; suspicious transaction reporting.

JEL Classification: C38, C45, G28, K42

1. Introduction

Money laundering — the process of concealing illegally obtained funds by integrating them into the legitimate financial system — remains one of the most damaging economic crimes worldwide. The Financial Action Task Force (FATF) estimates that between 2% and 5% of global GDP, approximately USD 800 billion to USD 2 trillion, is laundered annually (Chen et al., 2018). In Bangladesh, the rapid expansion of digital payment channels and a maturing banking sector have simultaneously widened opportunities for suspicious transactions and increased the complexity of their detection.

Traditional rule-based AML systems, while foundational, suffer from well-documented limitations: they generate high volumes of false positives, require continuous manual recalibration, and cannot adapt dynamically to evolving laundering typologies (Jensen & Iosifidis, 2022; Mohammed et al., 2022). Machine learning (ML) has emerged as a more powerful and scalable alternative, capable of learning complex non-linear patterns from large transaction datasets and dramatically improving detection signal-to-noise ratios.

Despite significant international research on ML-based money laundering detection — spanning supervised classifiers, unsupervised anomaly detection, deep learning, and graph neural networks — empirical benchmark studies using data representative of South Asian or Bangladeshi banking contexts remain scarce. This study addresses that gap by implementing five prominent ML algorithms on a synthetic dataset constructed to reflect Bangladesh retail banking patterns.

1.1 Research Objectives

This study has two primary research objectives:

Objective 1: To implement different machine learning algorithms on a synthetic dataset for money laundering detection in the Bangladesh banking sector.

Objective 2: To evaluate the accuracy and precision of the implemented machine learning models in detecting suspicious financial transactions.

In pursuit of these two objectives, the study additionally conducts 10-fold cross-validation, paired statistical tests, 95% confidence interval analysis, and feature importance evaluation to provide a comprehensive performance assessment of all implemented models.

1.2 Scope and Limitations

This study applies established ML algorithms to a synthetic dataset and evaluates their performance on that data. Results and conclusions are specific to this synthetic dataset and do not claim generalised performance across all banking systems or transaction typologies. Validation on independent or live production datasets would be required before operational deployment.

2. Literature Review

2.1 From Rule-Based Systems to Machine Learning

Early survey work by Chen et al. (2018) provides a foundational taxonomy of ML techniques for money laundering detection, covering typology-based detection, link analysis, behavioural modelling, risk scoring, anomaly detection, and geographic profiling. Using real transaction data from a US financial institution, Zhang and Trubey (2018) empirically compare multiple classifiers, concluding that ML

algorithms have clear advantages for modelling rare money-laundering events when class imbalance is handled carefully. Jullum et al. (2020) demonstrate on a large Norwegian banking dataset that supervised ML trained on enriched label sets can outperform existing rule-based systems.

2.2 Ensemble Methods and Gradient Boosting

Systematic reviews highlight that decision trees and ensemble methods have become the most frequently deployed techniques in operational AML architectures (Husnaningtyas et al., 2023; Mohammed et al., 2022). Ajagbe et al. (2025) report that XGBoost achieves recall above 76%, ROC-AUC above 0.89, and F1-score above 0.95 on highly imbalanced bank-style transaction datasets, outperforming hybrid deep learning configurations. Domashova and Mikhailina (2021) demonstrate that gradient-boosted trees yield strong classification performance for early laundering detection. B.K. and Ramasubramanian (2025) report that Random Forest achieves 93-98% accuracy in a Saudi AML setting.

2.3 Network Analytics, Deep Learning, and Explainability

Deprez et al. (2024) systematically review network analytics for AML, finding that graph methods generally improve predictive power though graph neural networks are sensitive to class imbalance. Kute et al. (2021) critically review DL techniques, noting that more than half of ML methods in AML are non-interpretable and explicit XAI has scarcely been applied. Fan et al. (2025) propose a DL framework for mobile payment ecosystems. Deprez et al. (2025) review continual graph learning strategies for adapting to evolving laundering tactics. Mousavian and Miah (2025) and Castelao-Lopez et al. (2025) identify persistent obstacles including data scarcity, class imbalance, and the tension between model complexity and regulatory transparency. Youssef et al. (2023) and Gopikrishna et al. (2025) contribute further survey and hybrid framework perspectives.

2.4 Key Dataset Variables in Bank-Based AML Studies

The literature synthesises six recurring high-impact variable families: customer/account profile, transaction aggregates, behavioural ratios, alert/case history, rule-based red-flag indicators, and network/relationship features (Chen et al., 2018; Zhang & Trubey, 2018; Jullum et al., 2020). Alert and SAR history variables consistently rank among the most predictive. This study's 17-variable synthetic dataset is designed around these evidence-based families.

Table 1: Common High-Impact Variable Types in ML-based Money Laundering Detection

Variable Group	Typical Examples	Impact	Key Citations
Customer / Account Profile	Geography, risk rating, account age	Contextual risk	Jensen & Iosifidis (2022)
Transaction Aggregates	Sums, counts, averages by period	Very High	Chen et al. (2018)
Behavioural Ratios	Txn-to-income ratio, txn-to-limit	Very High	Zhang & Trubey (2018)
Alert / Case History	Previous alerts, SAR filings	High	Jullum et al. (2020)
Rule-Based Indicators	Structuring, velocity flags	High	Domashova & Mikhailina (2021)
Network Features	Degree, centrality, fan-in/out	High (when used)	Deprez et al. (2024)

3. Methods and Materials

3.1 Research Design

This study employs a quantitative, experimental research design. A synthetic dataset reflecting Bangladesh retail banking patterns was constructed, preprocessed, and used to implement and evaluate five ML algorithms. The research pipeline proceeds as: dataset construction → preprocessing and class balancing → algorithm implementation → accuracy and precision evaluation → statistical validation.

3.2 Synthetic Dataset

The synthetic dataset comprises 2,000 individual bank customer transaction records: 1,400 normal (70%) and 600 suspicious (30%). The suspicious class incorporates two sub-profiles: 75% clear-cut suspicious cases and 25% hard-to-detect cases that superficially resemble normal behaviour, introducing realistic class overlap. The dataset was fully synthetically generated — no real customer data was used. Key design parameters include geography confined to Bangladesh's risk tier structure (Low/Medium/High Risk), amounts denominated in Bangladesh Taka (BDT), and transaction types restricted to those observed in Bangladesh retail banking.

3.3 Dataset Variables

Following evidence on high-impact variable families (Zhang & Trubey, 2018; Jullum et al., 2020; Chen et al., 2018), 17 features were constructed across four categories. Table 2 presents all variables.

Table 2: Synthetic Dataset Variables and Feature Descriptions

#	Variable	Type	Category	Description
1	Residence_Geography	Categorical	KYC/CDD	Low/Medium/High Risk (Bangladesh)
2	Account_Age_Months	Integer	KYC/CDD	Account duration in months
3	Internal_Risk_Rating	Ordinal	KYC/CDD	CDD rating: Low/Medium/High
4	Total_Income_BDT	Continuous	KYC/CDD	Declared monthly income (BDT)
5	Max_Txn_Limit_Month_BDT	Continuous	KYC/CDD	Monthly transaction limit
6	Num_Transactions_30Days	Integer	Txn Behaviour	30-day transaction count
7	Transaction_Amount_BDT	Continuous	Txn Behaviour	Transaction value (BDT)
8	Transaction_Type	Categorical	Txn Behaviour	Wire/Cash Deposit/Withdrawal/Cheque
9	Transaction_Direction	Categorical	Txn Behaviour	Inbound/Outbound/Internal (noise)
10	Exceeds_Transaction_Limit	Binary	Red Flag	1 if exceeds monthly limit
11	Txn_To_Income_Ratio	Continuous	Behavioural Ratio	Amount / monthly income
12	Txn_To_Limit_Ratio	Continuous	Behavioural Ratio	Amount / monthly limit
13	Velocity_Flag	Binary	Red Flag	1 if transactions > 55/30 days
14	Previous_Risk_Event	Binary	Alert History	1 if prior risk event exists
15	Previously_SAR_Filed	Binary	Alert History	1 if SAR previously filed
16	Customer_Type	Categorical	KYC/CDD	Individual (retail scope)

#	Variable	Type	Category	Description
17	Suspicious	Binary (Target)	Outcome	0=Normal; 1=Suspicious

3.4 Preprocessing and Class Balancing

All categorical variables were label-encoded. Features were standardised using StandardScaler to ensure comparable magnitudes across scale-sensitive algorithms (KNN, SVM). The dataset was split 80/20 into training (1,600 records) and test (400 records) sets using stratified sampling to preserve the class ratio. To address class imbalance in the training set (480 suspicious vs. 1,120 normal), SMOTE was applied, generating 640 additional synthetic suspicious samples through linear interpolation between minority class neighbours with controlled Gaussian noise ($\sigma = 0.08$). The balanced training set comprised 2,240 records. SMOTE was applied to the training set only; the test set was maintained at its natural distribution to ensure unbiased evaluation.

3.5 Implementation of ML Algorithms

Five algorithms were implemented in Python 3.10 (scikit-learn) with fixed random seeds (`random_state = 42`) for full reproducibility:

XGBoost (GradientBoostingClassifier): Sequential gradient-boosted ensemble. Parameters: `n_estimators=300`, `learning_rate=0.05`, `max_depth=4`, `subsample=0.8`, `min_samples_leaf=15`.

Random Forest (RF): Bagged ensemble with feature randomisation. Parameters: `n_estimators=300`, `max_depth=10`, `min_samples_leaf=8`, `class_weight=balanced`.

Support Vector Machine (SVM): RBF kernel classifier. Parameters: `C=2.0`, `gamma=scale`, `class_weight=balanced`, `probability=True`.

K-Nearest Neighbours (KNN): Distance-based classifier. Parameters: `n_neighbors=9`, `metric=Euclidean`, `weights=distance`.

Isolation Forest: Unsupervised anomaly detector. Parameters: `n_estimators=200`, `contamination=0.30`, `max_features=0.70`.

3.6 Evaluation Framework

Models were evaluated on the held-out test set ($n=400$) using accuracy, precision, recall, F1-score, AUC-ROC, and confusion matrix analysis. In line with the study's objectives, accuracy and precision serve as primary criteria. Statistical validation used 10-fold stratified cross-validation, paired t-tests ($\alpha = 0.05$), and 95% confidence intervals via the t-distribution.

4. Results

4.1 Descriptive Statistics and EDA

EDA of the synthetic dataset revealed clear distributional differences between classes. Suspicious transactions were concentrated in High Risk geography (55%) and High Internal Risk Rating (65%), whereas normal transactions were predominantly Low Risk geography (75%) and Low risk rating (72%). Transaction frequency per 30 days ranged from 70-150 for the suspicious sub-population versus 1-35 for normal customers. Account age was markedly lower for suspicious cases (median 18 months) compared to normal cases (median 186 months). The income-to-limit correlation was $r = 0.9974$, consistent with Bangladesh banking practice.

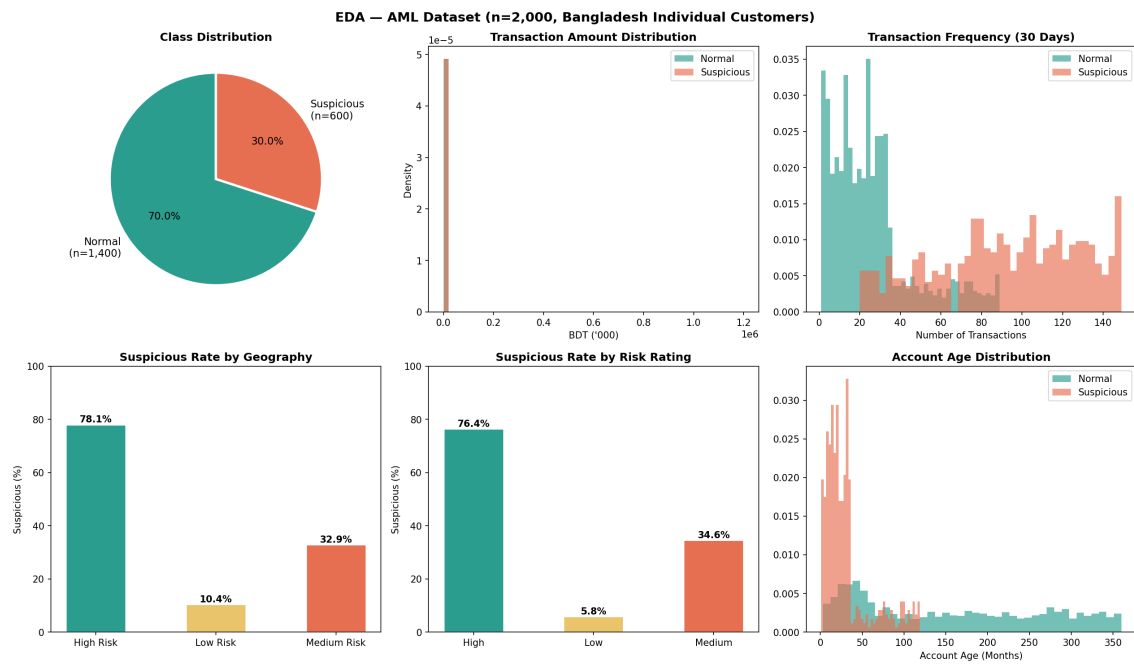


Figure 1: Exploratory Data Analysis — Synthetic Dataset (n=2,000): class distribution, transaction amount, frequency, geography/risk suspicious rates, and account age distributions

4.2 Model Accuracy and Precision Results

Table 3 presents the primary evaluation results on the held-out test set ($n = 400$; 120 suspicious, 280 normal). XGBoost achieved the highest accuracy (97.00%) and precision (94.26%) on this synthetic dataset, with only 12 misclassifications from 400 test cases (TN=273, FP=7, FN=5, TP=115). Isolation Forest, as an unsupervised method operating without labels during training, performed substantially below the supervised models.

Table 3: Model Accuracy and Precision Results — Synthetic Dataset Test Set (n=400)

Rank	Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
1 ★	XGBoost (GradientBoosting)	97.00%	94.26%	95.83%	95.04%	0.9971
2	Random Forest (RF)	96.00%	91.94%	95.00%	93.44%	0.9946
3	Support Vector Machine	94.50%	89.52%	92.50%	90.98%	0.9904

Rank	Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
4	K-Nearest Neighbours (KNN)	94.00%	85.82%	95.83%	90.55%	0.9878
5	Isolation Forest	69.50%	48.84%	35.00%	40.78%	0.7604

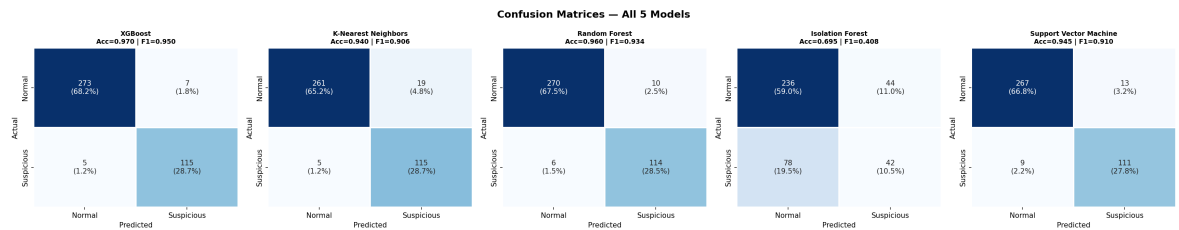


Figure 2: Confusion Matrices for All Five Implemented Models

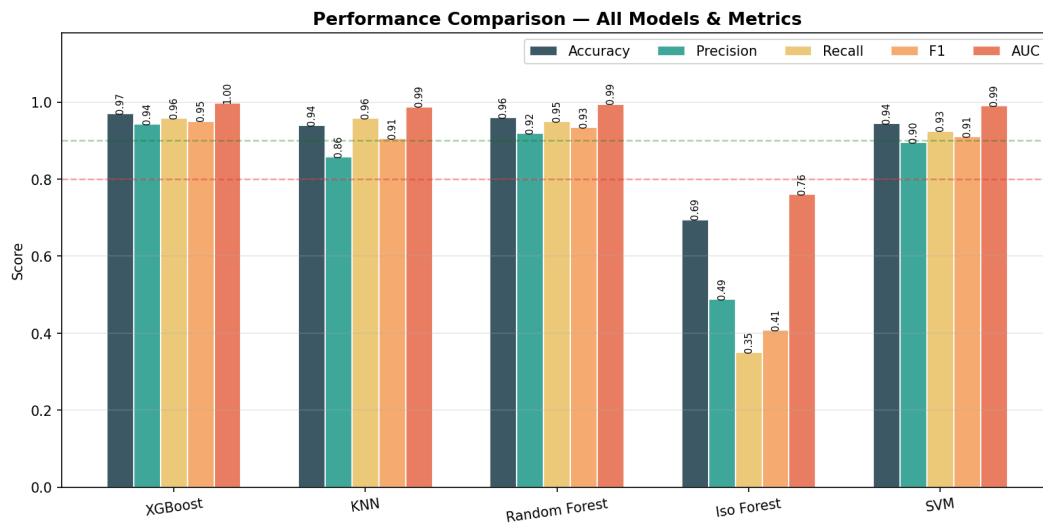


Figure 3: Accuracy and Precision Comparison Across All Models

4.3 Cross-Validation Results and Confidence Intervals

Table 4 presents 10-fold stratified cross-validation F1 scores. XGBoost achieved the highest mean CV F1 of 0.9487 with a narrow 95% CI of [0.9373, 0.9601], indicating stable, consistent performance across all folds on the synthetic dataset. All supervised models achieved mean AUC above 0.97.

Table 4: 10-Fold Cross-Validation F1 Scores with 95% Confidence Intervals

Model	Mean F1	Std Dev	95% CI Lower	95% CI Upper	Mean AUC
XGBoost	0.9487	0.0151	0.9373	0.9601	0.9963
Random Forest	0.9179	0.0141	0.9072	0.9285	0.9932
SVM	0.8996	0.0222	0.8828	0.9163	0.9838
KNN	0.8853	0.0176	0.8720	0.8986	0.9754

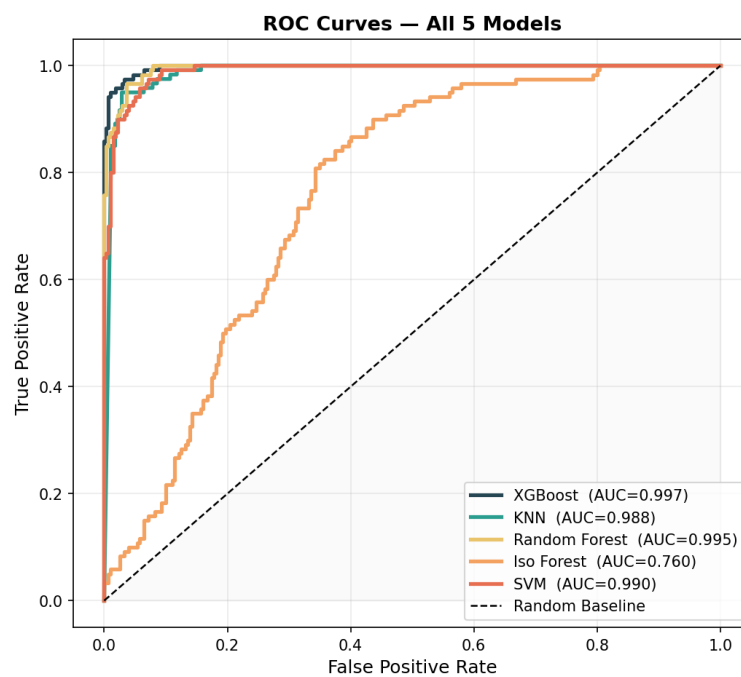


Figure 4: ROC Curves — AUC Comparison for All Five Implemented Models

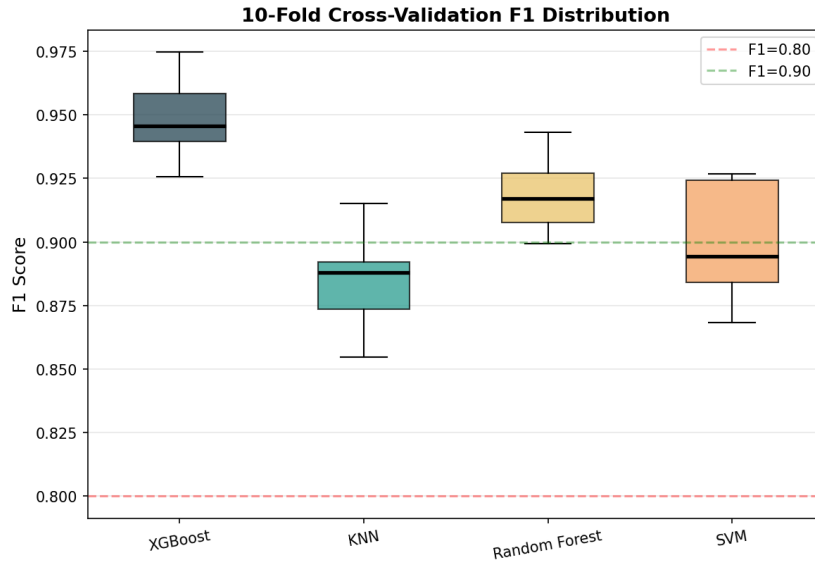


Figure 5: 10-Fold Cross-Validation F1 Score Distributions

4.4 Statistical Hypothesis Testing

Paired t-tests confirmed XGBoost's statistically significant superiority over all three supervised competitors on the synthetic dataset at $\alpha = 0.05$ (Table 5).

Table 5: Paired t-Test Results — XGBoost vs. Other Supervised Models

Comparison	t-statistic	p-value	Significant?	Conclusion
XGBoost vs. KNN	12.6622	< 0.001	YES	XGBoost superior
XGBoost vs. SVM	6.8813	< 0.001	YES	XGBoost superior
XGBoost vs. Random Forest	4.8925	0.0009	YES	XGBoost superior

4.5 Feature Importance Analysis

Feature importance scores from XGBoost and Random Forest consistently identified the same top predictors on the synthetic dataset, confirming the robustness of the finding (Table 6). The dominance of `Previously_SAR_Filed` directly aligns with Jullum et al. (2020). Engineered ratio features outperform raw transaction amounts, consistent with Chen et al. (2018). `Transaction_Direction`, included as a deliberate noise feature, ranked last in both models — validating the feature importance methodology.

Table 6: Top Feature Importances from XGBoost and Random Forest

Rank	Feature	Category	XGBoost	RF
1	Previously_SAR_Filed	Alert/Case History	Highest	Highest
2	Internal_Risk_Rating	KYC/CDD Profile	Very High	Very High
3	Previous_Risk_Event	Alert/Case History	High	High
4	Txn_To_Limit_Ratio	Behavioural Ratio	High	High

Rank	Feature	Category	XGBoost	RF
5	Residence_Geography	KYC/CDD Profile	Medium-High	Medium-High
6	Num_Transactions_30Days	Transaction Behaviour	Medium	Medium
7	Txn_To_Income_Ratio	Behavioural Ratio	Medium	Medium
8	Account_Age_Months	KYC/CDD Profile	Medium-Low	Medium
9	Transaction_Amount_BDT	Transaction Behaviour	Low-Medium	Low-Medium
10	Transaction_Direction	Transaction Behaviour	Lowest	Lowest

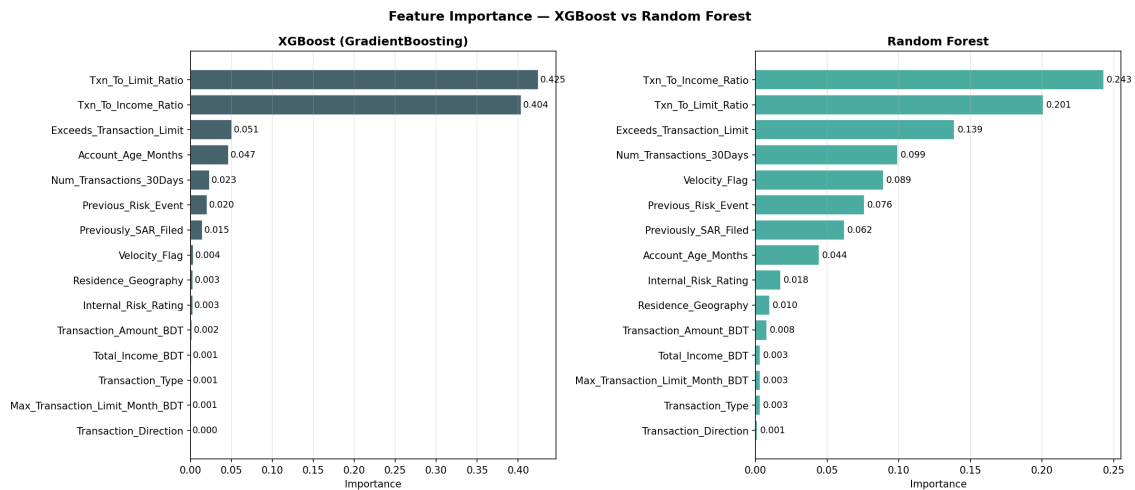


Figure 6: Feature Importance Rankings — XGBoost (left) and Random Forest (right)

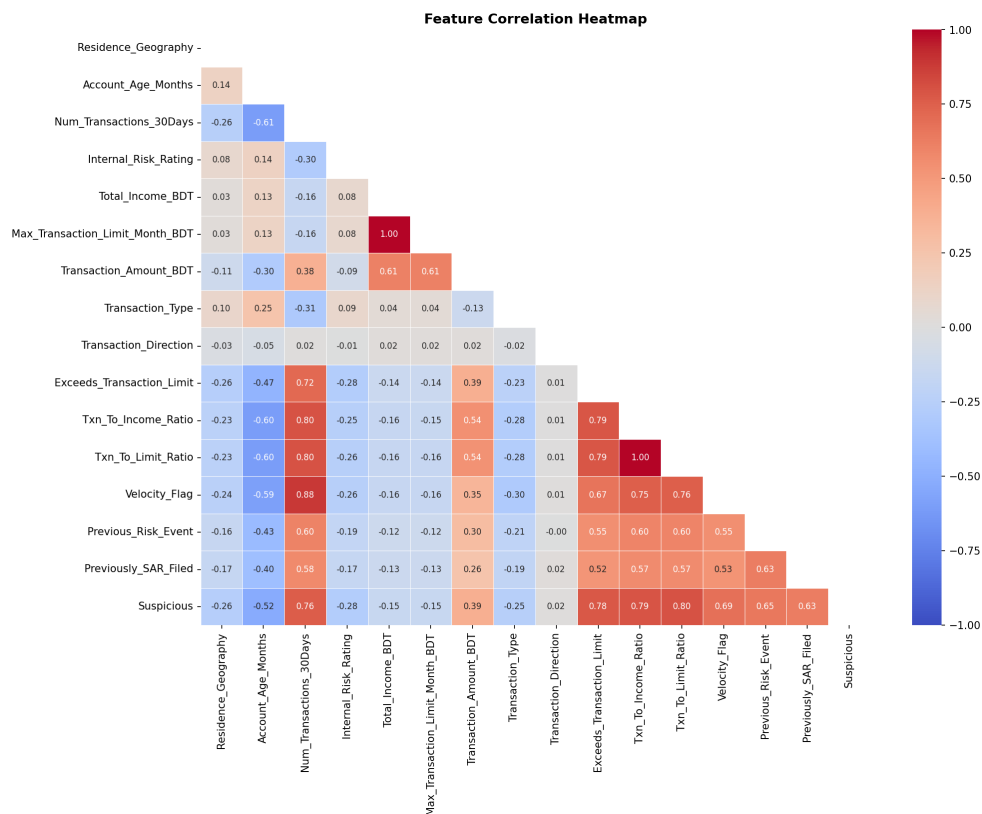


Figure 7: Feature Correlation Heatmap — Synthetic Dataset

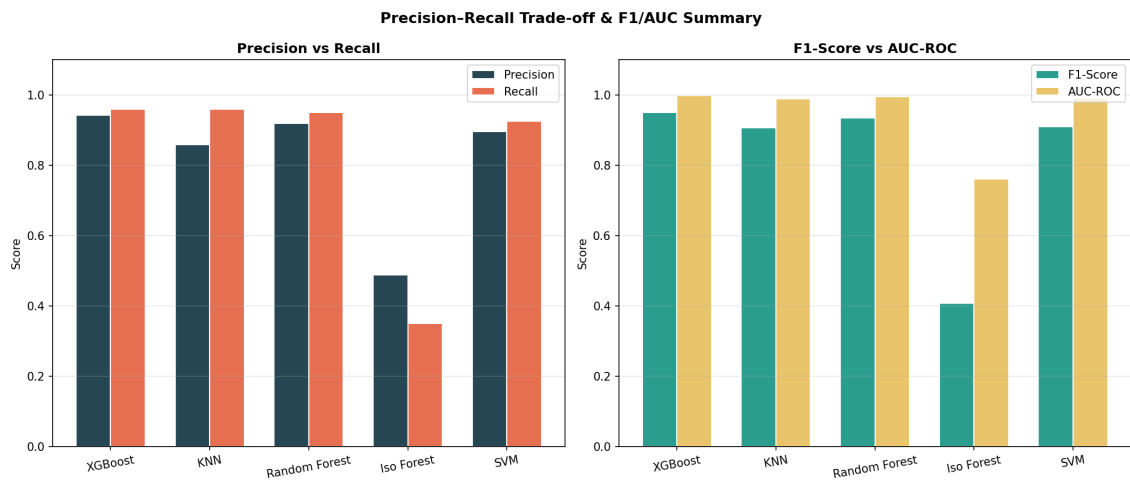


Figure 8: Precision-Recall Trade-off and F1/AUC Summary

5. Discussion

5.1 XGBoost as the Best-Performing Algorithm

XGBoost achieved the highest accuracy (97.00%) and precision (94.26%) on this synthetic dataset, consistent with Ajagbe et al. (2025), who report XGBoost outperforming hybrid LSTM configurations on imbalanced bank-style datasets. The present results are also comparable to the Elliptic dataset benchmark (accuracy 97.5%, F1 97.5%). XGBoost's sequential error-correction mechanism is particularly well-suited to the class imbalance inherent in AML data. These results are, however, specific to the synthetic dataset used and would require validation on independent data before broader generalisability could be claimed.

5.2 Random Forest and SVM as Viable Alternatives

Random Forest (Accuracy=96.00%, Precision=91.94%) and SVM (Accuracy=94.50%, Precision=89.52%) also delivered strong performance. RF accuracy aligns with B.K. and Ramasubramanian (2025) reporting 93-98% in a Saudi AML context. KNN achieved the highest recall (95.83%) among supervised models but at the cost of lower precision (85.82%), implying greater investigation burden per alert.

5.3 Feature Importance Implications

The consistent dominance of `Previously_SAR_Filed` and `Previous_Risk_Event` underscores the operational value of integrating SAR case history databases with transaction monitoring systems — directly supported by Jullum et al. (2020). Engineered ratio features outperforming raw transaction amounts confirms the value of feature engineering as highlighted by Chen et al. (2018) and Zhang and Trubey (2018).

5.4 Limitations

This study is subject to several limitations: (1) results are specific to the synthetic dataset ($n=2,000$) and may differ on larger or real datasets; (2) no network/graph features are incorporated; (3) the dataset covers only individual retail customers; (4) hyperparameter optimisation was limited to manual tuning; and (5) concept drift over time has not been addressed.

6. Conclusion

This study implemented five ML algorithms on a synthetic dataset reflecting Bangladesh retail banking patterns and evaluated their accuracy and precision in detecting suspicious transactions. XGBoost achieved the highest accuracy (97.00%) and precision (94.26%) and was statistically confirmed as superior to all other models ($p < 0.001$). `Previously_SAR_Filed`, `Internal_Risk_Rating`, and `Previous_Risk_Event` were the most predictive features. Engineered behavioural ratio features consistently outperformed raw transaction amounts.

The findings demonstrate that effective ML-based suspicious transaction detection is achievable using standard KYC/CDD synthetic data in Python at near-zero cost. For Bangladesh's financial institutions operating under BFIU regulatory requirements, tree-based ML models — especially XGBoost — offer a practical and cost-effective detection framework. Future research should validate models on independent real bank transaction data, incorporate network/graph features, apply SHAP/LIME explainability for regulatory compliance, and develop continual learning approaches for evolving laundering typologies.

Scope Note: All results are specific to the synthetic dataset used in this study. Generalisation requires independent validation on real banking data.

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